MOHD RASHID FAROOQUI

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EDUCATION

Indian Institute of Technology (IIT) Delhi – Abu Dhabi 08/2025 – 06/2027 (Expected)
M.Tech, Energy Transition & Sustainability —GPA 10 /10
(2 of 4 semesters completed)

Coursework:

- Process Engineering; Engineering Mathematics; Economics & Financing of Energy Transition; Energy Systems Modelling & Analysis; Decarbonizing the Fossil-Fuel Sector; AI in Energy systems; Energy, Development & Sustainability.
- **Laboratory:** Energy Systems & Computational Laboratory.

National Institute of Technology (NIT), Silchar, India 07/2008 – 06/2012
B.Tech, Mechanical Engineering — GPA 8.71 / 10

RESEARCH INTERESTS

My research sits at the intersection of energy-systems engineering, the energy transition, and the techno-economic assessment of low-carbon infrastructure. I am particularly interested in the **integrity-constrained repurposing of legacy oil and gas assets** — pipelines, terminals, and storage facilities — for hydrogen and CO_2 service, and in the **subsurface storage of hydrogen and CO_2 and CO_2-EOR**. My work combines **life-cycle assessment (LCA), techno-economic analysis (TEA), and data-driven / AI methods** to evaluate decarbonization pathways across the natural-gas, refining, and steel value chains. Drawing on more than a decade designing and engineering the very infrastructure the sector now seeks to decarbonize, I aim to pair hard-science rigor with the capital- and policy-systems perspective required to deploy these technologies at scale.

PUBLICATIONS

1. **Rashid Farooqui**, A. Agrawal, *"Hydrogen Systems & Infrastructure Integration,"* ASME International Mechanical Engineering Congress & Exposition (IMECE-India), 2025. doi:10.1115/IMECE-INDIA2025-159422.
2. **Mohd Rashid Farooqui**, *"Hydrogen: Powering India's Path to Net-Zero"* (cover article), *Energy Future*, The Energy and Resources Institute (TERI), Vol. 13, Jan to June Issue 3, 2025.
3. **Rashid Farooqui**, *"Clean Technologies for Energy Transition,"* Annual Technical Volume, Interdisciplinary Coordination Committee, The Institution of Engineers (India), 2025.

MANUSCRIPTS IN PREPARATION

1. **M. R. Farooqui**, A. Negi, T. S. Khan, M. A. Haider, *"Decarbonization of the Steel Industry: A GREET-Based Life-Cycle Assessment,"* in preparation.

RESEARCH EXPERIENCE

Graduate Research — IIT Delhi – Abu Dhabi 08/2025 – Present

- **GREET-based LCA & techno-economic study:** *Decarbonizing UAE Steelmaking through Green Hydrogen and Nuclear-Powered DRI–EAF*.
- **ML-based forecasting of Global Horizontal Irradiance (GHI)** for the Abu Dhabi region using NSRDB data (2017–2019).
- **Techno-economic evaluation** of a 100 MW grid-connected solar PV power plant.
- **Term paper:** *AI for Energy Efficiency: Applications, Case Studies, and Numerical Analysis*.

Applied Industry Research — MECON Limited 07/2012 – 08/2025

- **Hydrogen blending in the existing natural-gas grid:** structural-integrity and process-parameter analysis on industry data.
- **Seismic response of buried cross-country gas pipelines in fault zones:** built a CAESAR II model and mathematical framework from survey and design data, delivered in-house and eliminating outsourcing cost.

SPECIALIZED RESEARCH & TECHNICAL TRAINING

ADNOC Onshore, Abu Dhabi, UAE Summer 2026 (10 weeks)

- **Reservoir characterization & simulation:** Hands-on exposure to geology, geophysics, petrophysics, well logging, and SCAL through an EOR reservoir-simulation project.
- **Petroleum Engineering Learning Program:** Drilling, rigless well operations, artificial lift (ESP design), flow assurance, and well stimulation including hydraulic fracturing.

PROFESSIONAL EXPERIENCE

MECON Limited — Govt. of India engineering & project-management consultancy 07/2012 – 08/2025
Delhi, India

Across 13 years I have designed and engineered the oil & gas infrastructure — cross-country pipelines, terminals, storage and gas networks

Selected projects (Design, Engineering & Project Management):

- LPG **cross-country pipeline & terminals** (~200 km), HPCL — Detail engineering and project management.
- Greenfield **multiproduct white-oil storage terminal** (200,000 kL), HPCL — Concept to commissioning.
- Cross-country oil pipeline & terminals (~250 km), Chattogram–Dhaka, Bangladesh Petroleum Corp. — FEED, detail engineering, and terminal modernization.
- **Hydrogen blending** in natural-gas pipeline, Avantika Gas — Feasibility, techno-economic, and regulatory gap analysis.
- Modernization & augmentation of the national **natural-gas grid** (pan-India), GAIL — network mapping, retrofitting, automation, and energy-efficiency optimization.
- **Vapor-recovery systems** at white-oil terminals (pan-India), HPCL — Emissions-control engineering and regulatory compliance.
- **Crude-oil storage tanks** & ancillary facilities, Oil India — Technical specifications, basic engineering, and tendering.
- **Iron-ore slurry cross-country pipeline** (techno-economic feasibility), NMDC — GIS route survey, basic engineering, and cost analysis.
- Leh, Ladakh strategic terminal: designed a white-oil multiproduct storage terminal at ~3,500 m AMSL and ~−30 °C for HPCL.

- HINCOL **bitumen** plant, Visakhapatnam: designed and commissioned the augmentation of the HPCL–COLAS (France) joint-venture plant.

TALKS & PRESENTATIONS

- "Hydrogen as a Green Fuel," Global Tech Conference, Delhi, India, 2024.

TEACHING

- **Teaching Assistant,** AMEP 1000 — Engineering Visualization, IIT Delhi – Abu Dhabi (Spring 2025).

COURSE CERTIFICATIONS

- **Energy within Environmental Constraints** — HarvardX (Prof. David Keith).
- **Energy Economics and Policy** — MITx (Prof. Christopher Knittel).
- **Probability: The Science of Uncertainty and Data** — MITx MicroMasters (Prof. John N. Tsitsiklis).
- **Energy Resources, Economics and Environment** — NPTEL (Prof. Rangan Banerjee).

TECHNICAL SKILLS

- **Modelling & simulation:** Aspen, ANSYS, GREET, CAESAR II, SAM, AutoCAD 3D, Material Studio
- **Programming & analysis:** Python, C / C++, MATLAB.
- **Methods:** Life-cycle assessment (LCA), Techno-economic analysis (TEA), FEED & detail engineering, system optimization, feasibility studies, machine learning for energy systems.

AWARDS & HONORS

- **Ranked 5th of 110,** B.Tech Mechanical Engineering, NIT Silchar.
- **Top 1%** (~1 million candidates), All-India Engineering Entrance Examination (AIEEE).
- **27th rank** (~2.5 million candidates), Higher Secondary Board Examination (Uttar Pradesh)
- **Silver medal,** All India inter NIT handball tournament.

SPORT & OTHER ACTIVITY

- **Professional sport:** Represented Institute and company for Badminton and Volleyball.
- **NGO "Blessings" (Member):** Child-education and women's-empowerment initiatives; taught and mentored children from marginalized communities.